Rishabh Patidar

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Education

Madhav Institute of Technology and Science — Gwalior, MP
B.Tech in Information Technology | ***CGPA: 8.124*** — *Expected: May 2027*
Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, DBMS, Statistics, Linear Algebra

Technical Skills

Languages: Python, SQL, Java, C/C++
AI / ML: Machine Learning, NLP, Retrieval-Augmented Generation (RAG), LLaMA-3.3-70B, Vector Embeddings, Prompt Engineering, Groq API, Pinecone, ChromaDB
ML Frameworks: Scikit-learn, XGBoost, Random Forest, Gradient Boosting, KMeans, PCA, SHAP, SMOTE, Feature Engineering, Hyperparameter Tuning, Statistical Analysis
Data & Tools: Pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy, FastAPI, Streamlit, Git, Docker (basic), Jupyter
Database: SQL (schema design, complex queries, ETL pipelines), MongoDB

Experience

Undergraduate Research Assistant — Jan – May 2026
Madhav Institute of Technology and Science — *Gwalior, MP*

- Developed a Python-based ML pipeline for event footfall prediction, conducting EDA on 50+ features with domain-informed feature engineering, reducing model noise by 30%
- Built a two-stage XGBoost Regressor ($R^2 = 0.96$) with a Dynamic Physical Clamping algorithm enforcing venue-capacity constraints at inference, achieving zero invalid predictions across all test cases
- Integrated LLaMA-3.3-70B via Groq API to develop an NLP layer for semantic brand-event synergy scoring, generating structured AI-driven negotiation reports deployed as a stateless FastAPI microservice

Projects

Code Archaeologist — Git History RAG System — *2026*
Python, NLP, RAG, Pinecone, BGE Embeddings, LLaMA-3.3-70B, Groq API, NetworkX, Streamlit

- Built a Python-based AI application for natural language querying of Git repositories — classifies developer intent via NLP (bug fix, feature, refactor, chore) and stores enriched chunks in Pinecone with per-repository namespace isolation
- Implemented hybrid retrieval combining dense semantic search (BGE embeddings) with BM25 statistical reranking at 60/40 weight; engineered a NetworkX knowledge graph for temporal data analysis surfaced through a multi-repo Streamlit dashboard with LLM-generated summaries

Customer Segmentation & Churn Prediction — *2025*
Python, Scikit-learn, XGBoost, SHAP, SMOTE, SQL, Streamlit

- Designed a modular SQL-to-Pandas data pipeline to engineer 22 RFM and behavioural features from 2.5M+ raw transactions, reducing computation time by 40%; applied KMeans (k=4) clustering and trained a Random Forest classifier (ROC-AUC 0.8793) with SMOTE oversampling
- Used SHAP statistical analysis to identify recency as the dominant churn driver, translating model outputs into a $233,917 revenue-at-risk estimate across 392 flagged households via an interactive Streamlit dashboard

CinemaIQ — Box Office Intelligence Platform — *2025*
Python, XGBoost, Scikit-learn, Streamlit, Plotly, Groq API

- Developed Python-based predictive models with 30+ domain-specific features; benchmarked 5 ML algorithms selecting XGBoost for lowest RMSE, and integrated LLaMA-3.3-70B to generate AI-powered natural language commentary deployed in a Streamlit dashboard with scenario simulation and ROI analytics

Video Target Identification System — *2025*
Python, InsightFace, MediaPipe, OpenCV, Streamlit

- Developed a Python-based biometric AI application fusing InsightFace 512-d face embeddings and MediaPipe 12-d pose/gait descriptors via weighted cosine similarity for person re-identification across large multi-camera CCTV datasets; deployed Streamlit forensic dashboard with automated evidence export

Certifications & Achievements

Mathematical Foundations of Machine Learning | NPTEL, IIT Madras | Score: 73/100 | 2026
1st Runner-Up — SSH '26 National Hackathon | SSH | Feb 2026
Top Performer in WebDev Track — Hacksagon | ABV IIITM Gwalior | Apr 2026
Finalist — ABV-IIITM Hackatron | ABV-IIITM (Powered by GitHub) | Oct 2025